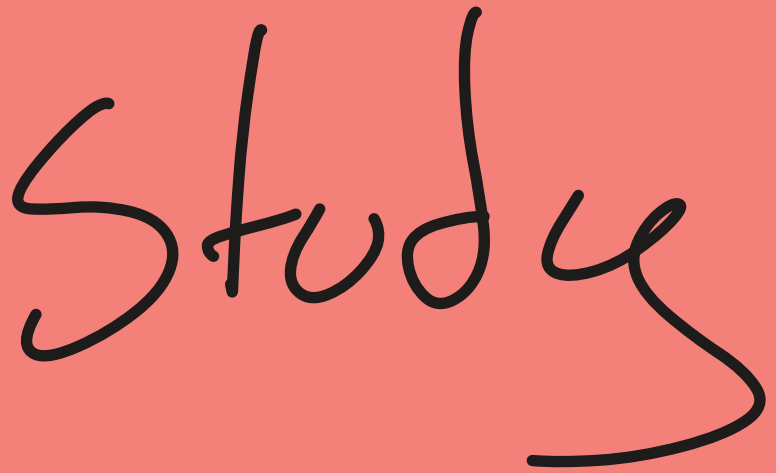

Study

travling at 105km/h for duration of 1h 50min

traffic slows you down to 70km/h. how long will it take

$v = \frac{d}{t}$

$d = v \cdot t$

$= (105\,km/h)(1.833h)$

$= 192.4965\,km$

$t = \frac{d}{v}$

$= \frac{192.4965km}{70km/h}$

$= 2.74995h$

$= 2h\ 45min$

$\therefore$ 55min Longer drive

read question carfully...

it ask how much Longer will it take to drive

A car stopped at traffic light

distance from the light $x(t) = bt^2 - ct^3$, where $b = 2.40 m/s^2$, $c = 0.120 m/s^3$

a) Calculate V_{avg} $t=0$ to $t=10$

$V(t) = 2bt - 3ct^2$

$V(10-0) = 2b(10) - 3c(10)^2$

$= 48 - 36$

$= 12 m/s$

b) find instant V at $t = 0, 5, 10$

$V(0) = 2bt - 3ct^2$

$= 0 m/s$

$V(5) = 2bt - 3ct^2$

$= 15 m/s$

$V(10) = 2bt - 3ct^2$

$= 12 m/s$

C) When is the car at rest again

$0 = 2bt - 3ct^2$

$0 = 4.8t - 0.36t^2$

$0 = t(4.8 - 0.36t)$

$t = 0$

$0.36t = 4.8$

$t = 13.33s$

textbook says answer is $13.3 m/s$?

Speed?

Cars velocity: $V_x(t) = \alpha + \beta t^2$, where $\alpha = 3.00 m/s$ and $\beta = 0.100 m/s^3$

a) calculate avg velocity 0-5s

$V(5) = 5.50 m/s$

$V(0) = 3.00 m/s$

$$a = \frac{5.50 - 3.00}{5.00 - 0}$$

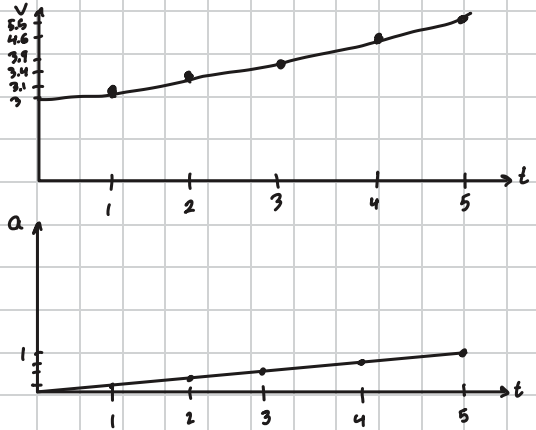
$$= 0.500 m/s^2$$

b) instant a at 0 and 5

$a(0) = 2(0.100)(0) = 0 m/s^2$

$a(5) = 2(0.100)(5) = 1 m/s^2$

c) graphs of v and a from 0-5s



$a(t) = 2(0.100)(t)$

$=$

1 -> 0.2
2 -> 0.4
3 -> 0.6
4 -> 0.8
5 -> 1

Object has cards $(1.1, 3.4)$ at $t_1 = 0$

$(5.3, -0.5)$ at $t_2 = 3.0\text{s}$

a) find the components of avg velocity

$\Delta d_y = y_2 - y_1 = -3.9$

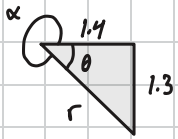
$\Delta d_x = x_2 - x_1 = 4.2$

$$V = \frac{\Delta d}{\Delta t}$$

$V_y = -1.3\text{m/s}$

$V_x = 1.4\text{m/s}$

b) magnitude and direction



$r = \sqrt{1.4^2 + 1.3^2}$

$= 1.9\text{m/s}$ ← magnitude

$\theta = \tan^{-1}\left(\frac{1.3}{1.4}\right)$

$= 43^\circ$ ← direction

$\alpha = 317^\circ$

Bird flying given by $x(t) = \alpha t$ and $y(t) = 3.0m - \beta t^2$

where $\alpha = 2.4 m/s$ and $\beta = 1.2 m/s^2$

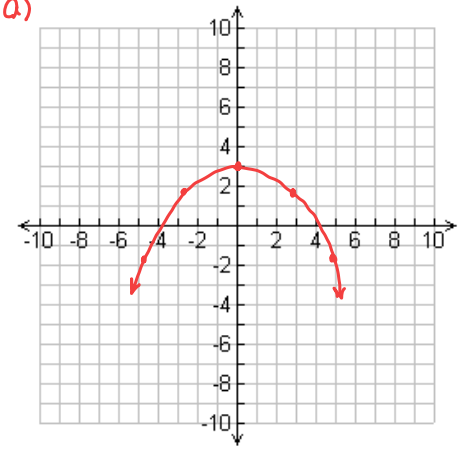
a) Sketch path from 1 – 2s

$x(0) = 2.4(0) = 0$ $\quad y(0) = 3.0m - (1.2)(0)^2 = 3.0$

$x(1) = 2.4(1) = 2.4$ $\quad y(1) = 3.0m - (1.2)(1)^2 = 1.8$

$x(2) = 2.4(2) = 4.8$ $\quad y(2) = 3.0m - (1.2)(2)^2 = -1.8$

a)



b) find vector for $\vec{V}$ and $\vec{a}$ as function of time

$V(t) = V_x(t)\hat{i} + V_y(t)\hat{j}$ $\qquad a(t) = a_x(t)\hat{i} + a_y(t)\hat{j}$

$V_x(t) = \alpha$ $\qquad a_x(t) = 0$

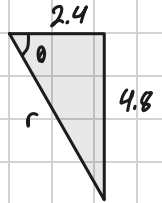
$V_y(t) = -2\beta t$ $\qquad a_y(t) = -2\beta$

$\boxed{V(t) = (2.4\hat{i} - 2\beta t\hat{j})\, m/s}$ $\qquad \boxed{a(t) = -2\beta\hat{j}\, m/s^2}$

c) find mag and direction for $\vec{V}$ and $\vec{a}$ at $t = 2.0s$

$V_x(2) = \alpha = 2.4 m/s$ $\qquad a_x(2) = 0$

$V_y(2) = -2(1.2)(2) = -4.8 m/s$ $\qquad a_y(2) = -2(1.2) = -2.4 m/s^2$

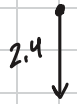


$r = \sqrt{2.4^2 + 4.8^2}$

$= \boxed{5.4 m/s}$

$\theta = \tan^{-1}\left(\frac{4.8}{2.4}\right)$

$= 63° \Rightarrow \boxed{297°}$

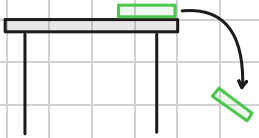


$\boxed{\text{mag} = 2.4 m/s^2}$

$\boxed{\theta = 270°}$

d) Speeding up, turning right

book slides off table, speed of 1.10 m/s. it hits the floor at 0.480 s



horizontal velocity is constant throughout

a) hight above floor

$$h = \cancel{v_i \Delta t^0} + \frac{1}{2} g \Delta t^2$$

$$= \frac{1}{2} g \Delta t^2$$

$$= \frac{1}{2}(9.80 m/s^2)(0.480 s)^2$$

$$= 1.13 m$$

b) horizontal distance

$$d = v_{ox} t$$

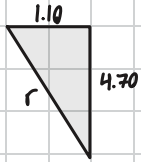
$$= (1.10 m/s)(0.480 s)$$

$$= 0.528 m$$

c) horizontal and vertical components befor hits ground

$$v_x = 1.10 m/s$$

$$v_y = -gt = -4.70 m/s$$



$$r = \sqrt{1.1^2 + 4.7^2}$$

$$= 4.83 m/s$$

$$\theta = \tan^{-1}\left(\frac{4.7}{1.1}\right)$$

$$= -76.8^\circ$$

$$= 283^\circ$$

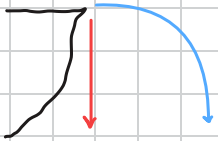
d)

two jump off a cliff

- Person 1 reaches the ground in 2.7s dropping vertically
- Person 2 jumps out with 95 cm/s

a) how far out will they jump

$95 \frac{cm}{s} \times \frac{1m}{100 cm} = 0.95 m/s$



$h = \frac{1}{2} g t^2$

$= \frac{1}{2}(9.8)(2.7)^2$

$= 35.721m$

Constant

$V_x = 0.95 m/s$

$V = \frac{d}{t}$

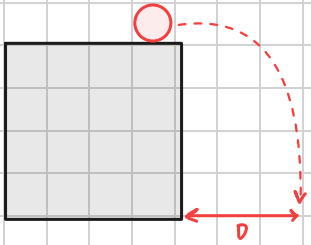
$d = v \cdot t$

$= (0.95 m/s)(2.70s)$

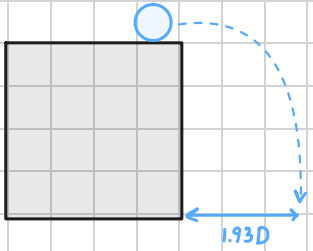
$= 2.57m$

we only look at horizontal numbers to find horizontal distance

ball drops off a table on earth



Same experament but on a different planet



a) whats the planets g?

V_x constant

$t = \sqrt{\frac{2h}{g}}$

$d = V_0 t$

$= V_0 \sqrt{\frac{2h}{g}}$

on earth $D = V_0 \sqrt{\frac{2h}{g_{earth}}}$

on planet $D = V_0 \sqrt{\frac{2h}{g_x}}$

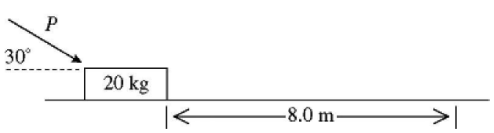
$$\frac{1.93D}{D} = \frac{\sqrt{\frac{2h}{g_x}}}{\sqrt{\frac{2h}{g_{earth}}}} \Rightarrow 1.93 = \sqrt{\frac{g_e}{g_x}}$$

$$1.93^2 = \frac{g_e}{g_x}$$

$$\frac{g_e}{1.93^2} = g_x$$

$$2.63 m/s^2 = g_x$$

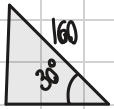
textbook says answer is $1.28 m/s^2$. GPT says wrong



$P = 160\text{N}$

$V_1 = 0.500\text{m/s}$

$V_2 = 2.60\text{m/s}$



$P_x = 160\cos 30$

$= 138.6\text{N}$

$W = (138.6)(8.00)$

$= 1108.8\text{ J}$

$W = Fd$

$E_k = \frac{1}{2}m(v_2^2 - v_1^2)$

$= \frac{1}{2}(20)(2.6^2 - 0.5^2)$

$= 65.1\text{ J}$

$W_{net} = W_p + W_f$

$65.1 = 1108.8 + W_f$

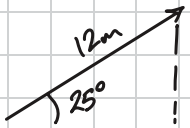
$W_f = -1043.7\text{ J}$

$\approx -1040\text{ J}$

55.8kg run up 12m hill incline 25° in 3s

$1\,hp = 746\,W$

$$P_{avg} = \frac{\Delta W}{\Delta t}$$



$$\Delta E = mg\Delta h$$
$$= (55.8kg)(9.8)(12\sin 25)$$
$$= 2776\,J$$

$$P = \frac{2776\,J}{3}$$
$$= 925.3\,W$$

$$hp = \frac{925.3}{746}$$
$$= 1.24\,hp$$

1000 kg 20 m/s $\mu = 0.8$ $\Delta d = ?$

$V_1^2 = V_2^2 + 2a\Delta d$

$F_{net} = ma$

$F_f = \mu N$

$ma = 0.8mg$

$a = 0.8(9.8)$

$\frac{V_1^2 - V_2^2}{2\Delta d} = a$

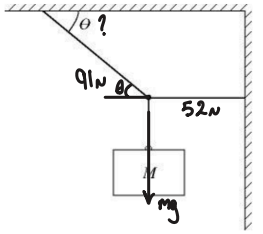
$\frac{V_1^2 - V_2^2}{2\Delta d} = 0.8(9.8)$

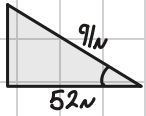
$\frac{-400}{2\Delta d} = 0.8(9.8)$

$\frac{-400}{2(0.8)(9.8)} = \Delta d$

$25.5 = \Delta d$

$26 \approx \Delta d$

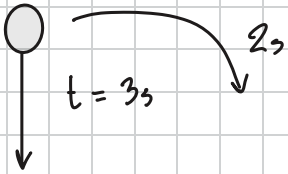




$\cos\theta = \left(\frac{52}{91}\right)$

$\theta = \cos^{-1}\left(\frac{52}{91}\right)$

$= 55°$



$h = \frac{1}{2}gt^2$

$= \frac{1}{2}(9.8)(3)^2$

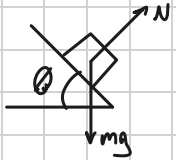
$= 44.1$

$d = V_0 t + \frac{1}{2}at^2$

$44.1 = V_0(2) + \frac{1}{2}(9.8)(2.0)^2$

$V_0 = 12.25 m/s$

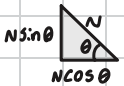
downwards



$m = 600$

$r = 110$

$V = 24.5$



$\cos\theta = \frac{adj}{N}$

$N\cos\theta = mg$ $\quad N\sin\theta = \frac{mv^2}{r}$

$\frac{N\sin\theta}{N\cos\theta} = \frac{\frac{mv^2}{r}}{mg}$

$\frac{N\sin\theta}{N\cos\theta} = \frac{v^2}{rg}$

$\tan\theta = \frac{v^2}{rg}$

$\theta = \tan^{-1}\left(\frac{v^2}{rg}\right)$

$= 29.1°$

1000kg car 20m/s stop $u = 0.8$

$V_1 = 20 m/s$

$V_2 = 0 m/s$

$m = 1000kg$

$u = 0.8$

$F_f = uN$

$ma = umg$

$a = (0.8)(9.8)$

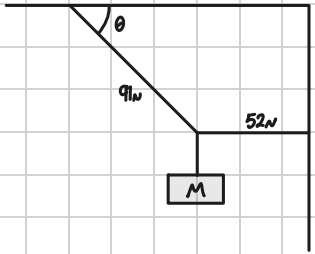
$= 7.84 m/s^2$

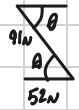
$V_2^2 = V_1^2 + 2ad$

$\frac{V_2^2 - V_1^2}{2a} = d$

$\frac{400}{15.68} = d$

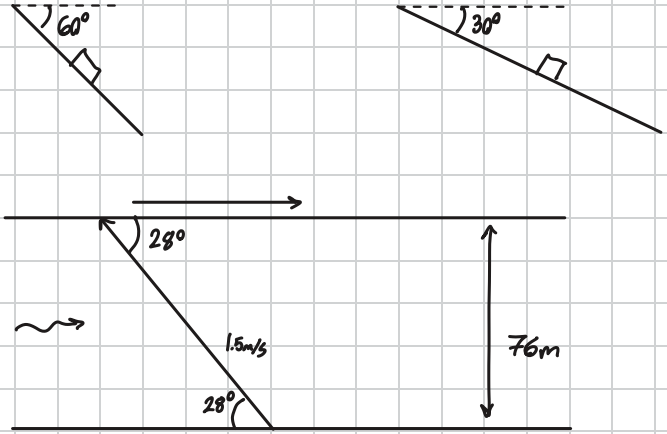
$25.5m = d$

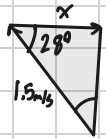




$\theta = \cos^{-1}\left(\frac{52}{91}\right)$

$= 55°$

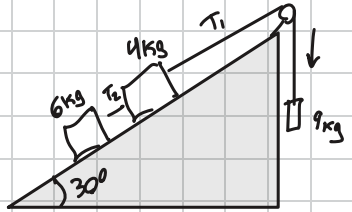




$\cos 28 = \frac{x}{1.5}$

$1.5\cos 28 = x$

$1.32\ m/s = x$



$F_{net} = 9g - (6g\sin 30) - (4g\sin 30)$

$= 88.2 - 29.4 - 19.6$

$= 39.2N$

$M_{Tot} = 6+4+9 = 19kg$

$a = \frac{39.2}{19} = 2.06 m/s^2$

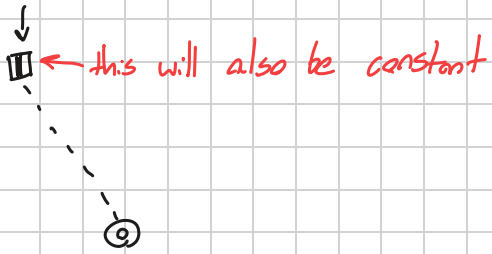
$T - m_1 g\sin 30 = m_1 a$

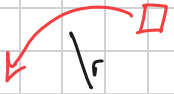
$T = m_1(g\sin 30 + a)$

$= 6(9.8\sin 30 + 2.06)$

$= 41.8N$







$V = 40 m/s$

$r = 200 m$

$u_s = ?$

$$F_f = F_c$$

$$u\cancel{mg} = \frac{\cancel{m}v^2}{r}$$

$$ug = \frac{v^2}{r}$$

$$u = \frac{(40 m/s)^2}{(200m)(9.8)}$$

$$= 0.816$$

$$2x + y - \sqrt{2}\sin(xy) = \frac{\pi}{2}$$

find the tangent equation @ $\left(\frac{\pi}{4}, 1\right)$

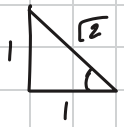
$$2 + \frac{dy}{dx} - \sqrt{2}\cos(xy)\cdot\left(x\frac{dy}{dx} + y\right) = 0$$

$$2 + \frac{dy}{dx} - \sqrt{2}\left(\frac{1}{\sqrt{2}}\right)\cdot\frac{dy}{dx} = 0$$

$$2 + \frac{dy}{dx} - 1\cdot\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} - \frac{dy}{dx} = \frac{-2}{\frac{\sqrt{2}}{\sqrt{2}}}$$

$$0 =$$



$$2x+y-\sqrt{2}\sin(xy)=\frac{\pi}{2} \quad \left(\frac{\pi}{4},1\right)$$

$$2+\frac{dy}{dx}-\sqrt{2}\cos(xy)\cdot x\frac{dy}{dx}+y=0$$

$$xy=\frac{\pi}{4}(1)=\frac{\pi}{4} \rightarrow \cos\left(\frac{\pi}{4}\right)=\frac{\sqrt{2}}{2}$$

$$2+\frac{dy}{dx}-\sqrt{2}\cos\left(\frac{\pi}{4}\right)\cdot\frac{\pi}{4}\frac{dy}{dx}+1=0$$